

PakPulse AI – AI & GIS Powered Multi-Disease Early Warning System

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Introduction & Problem Statement

Pakistan faces recurring outbreaks of multiple diseases such as dengue, malaria, cholera, influenza, and COVID-19. These outbreaks create seasonal and regional health crises, often overwhelming healthcare systems. Current responses are reactive, beginning only after hospital cases rise significantly. This delay increases costs, mortality, and strain on hospitals.

There is a clear need for a **multi-disease, data-driven early warning system** that predicts potential outbreaks, visualizes them geographically, and communicates risk to authorities in time to prevent large-scale crises.

Objectives

The proposed project will:

- Predict outbreak risks for **multiple diseases** (dengue, malaria, cholera, influenza, etc.) using climate, demographic, and historical case data.
- Visualize outbreak risks at the district level through an interactive GIS-enabled dashboard.
- Provide **disease-specific alerts** (via email/SMS) to relevant stakeholders.
- Deliver a working prototype that can later be expanded into a **national health surveillance system**.

Key Features

1. Multi-Disease Machine Learning Forecasting

- Predict outbreak probabilities for multiple diseases simultaneously.
- ML algorithms: Random Forest, XGBoost, LSTM for time-series predictions.

2. GIS Visualization

- Heatmaps of outbreak risks by district.
- Filters by disease type, date range, and location.
- Combined map showing multiple disease layers.

3. Alerts System

- Automatic email/SMS alerts when disease-specific risks cross thresholds.
- Multi-disease risk prioritization for districts facing multiple threats.

4. Anomaly Detection

- Detect sudden spikes across multiple diseases using Isolation Forest or DBSCAN.

5. Weather–Disease Correlation Analyzer

- Show which climatic or environmental factors drive outbreaks of different diseases.
- Example: rainfall → dengue/malaria; poor sanitation → cholera.

6. Risk Index Generator

- **Disease Risk Index (DRI)** for each disease (0–100).
- **Composite Health Risk Score** per district for overall vulnerability.

7. Explainable AI Module

- SHAP/LIME visualizations to explain which features influenced predictions.
- Enhances transparency and trust in AI recommendations.

8. Social Media Sentiment Mining (Optional)

- Scrape and analyze public posts in English/Urdu mentioning symptoms.
- Detect early community signals of outbreaks.

9. AI-Powered Q&A Chatbot (Optional)

- Streamlit/Gradio chatbot:
 “Which districts are at highest risk for malaria this month?”
 - Improves usability for non-technical stakeholders.
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Methodology

- **Data Collection:** Multiple disease datasets (NIH, WHO, district health departments), climate (OpenWeatherMap), demographics, sanitation, mobility, and social media.
 - **Preprocessing:** Cleaning, normalization, tagging by disease.
 - **Model Development:** Multi-disease forecasting, anomaly detection, correlation analysis, risk scoring.
 - **Evaluation:** Accuracy, RMSE, F1-score (for classification), anomaly precision.
 - **Dashboard:** Streamlit + Folium/Plotly visualizations.
 - **Alerts:** Disease-specific email/SMS notifications.
 - **Optional Modules:** Sentiment mining, chatbot, explainable AI integration.
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Scope

In Scope:

- Prediction of multiple disease outbreaks (dengue, malaria, cholera, influenza).
- GIS dashboard with multi-disease layers.
- Email/SMS alerts.
- Risk scoring + anomaly detection.

Out of Scope (Future Work):

- Blockchain for tamper-proof outbreak logs.
 - Community-driven reporting portals.
 - Hospital demand forecasting.
 - Mobile app deployment.
 - Nationwide, real-time system integration.
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Expected Outcomes

- A functional AI model for **multi-disease outbreak prediction**.

- A GIS dashboard visualizing risk by disease and location.
 - Disease-specific risk indices + alerts.
 - A prototype scalable to national use.
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Tools & Technologies

- **Programming:** Python
 - **Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, SHAP, LIME, Folium, Plotly
 - **Dashboard:** Streamlit
 - **Alerts:** Gmail SMTP / Twilio SMS API
 - **Data Sources:** NIH Pakistan, WHO, OpenWeatherMap, district health departments, social media APIs
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Risks & Mitigation

- **Data Scarcity:** Use synthetic/public data if real datasets are incomplete.
 - **Model Complexity:** Start with simpler classifiers, scale up later.
 - **Deployment Constraints:** Host locally or on Streamlit Cloud for demonstration.
 - **Social Media Noise:** Apply filters and lightweight NLP to reduce irrelevant chatter.
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Conclusion & Future Work

This project will deliver a **multi-disease, AI + GIS powered early warning system** for Pakistan, extending beyond dengue to include malaria, cholera, influenza, and other outbreaks. The system will enhance health authorities' ability to act proactively, reduce burden on hospitals, and save lives.

Future extensions may include blockchain for secure records, hospital demand forecasting, and mobile app development for broader accessibility.
